

MAT 102: Ordinary Differential Equations

Topic 4 — Laplace Transforms and Applications to Differential Equations

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What You Will Learn in This Topic

By the end of this topic, you should be able to:

1. Explain what an improper integral is and evaluate basic ones
2. Define the Laplace transform and compute it from the definition
3. Use standard transform tables and theorems to find transforms quickly
4. Find inverse Laplace transforms using partial fractions
5. Apply Laplace transforms step-by-step to solve ODEs with initial conditions
6. Solve systems of simultaneous differential equations using Laplace transforms

1 Background Ideas

Before we can understand the Laplace transform, we need to be comfortable with two things used directly in its definition: **improper integrals** and **integration by parts**.

1.1 What Is an Improper Integral?

In ordinary integration we compute $\int_a^b f(t) dt$ where both limits are **finite**. An **improper integral** is one where one or both limits is **infinity**, for example:

$$\int_0^{\infty} e^{-2t} dt$$

Since ∞ is not a real number we cannot substitute it directly. Instead we use a limit.

Definition: Improper Integral with Upper Limit ∞

$$\int_0^{\infty} f(t) dt = \lim_{T \rightarrow \infty} \int_0^T f(t) dt$$

- If the limit is a **finite number**, the integral **converges** to that number.
- If the limit is $\pm\infty$ or does not exist, the integral **diverges**.

Example 1: Evaluate $\int_0^{\infty} e^{-2t} dt$.

Step 1 — Replace ∞ with T and integrate:

$$\int_0^T e^{-2t} dt = \left[\frac{e^{-2t}}{-2} \right]_0^T = -\frac{e^{-2T}}{2} + \frac{1}{2}$$

Step 2 — Take the limit (since $s > 0$, $e^{-2T} \rightarrow 0$ as $T \rightarrow \infty$):

$$\lim_{T \rightarrow \infty} \left(-\frac{e^{-2T}}{2} + \frac{1}{2} \right) = 0 + \frac{1}{2}$$

$$\boxed{\int_0^\infty e^{-2t} dt = \frac{1}{2}}$$

Example 2: Evaluate $\int_0^\infty e^{3t} dt$.

$$\int_0^T e^{3t} dt = \frac{e^{3T}}{3} - \frac{1}{3} \xrightarrow{T \rightarrow \infty} \infty$$

$$\boxed{\int_0^\infty e^{3t} dt \text{ diverges}}$$

 **Tip**

Key pattern:

- Negative exponent e^{-at} , $a > 0 \Rightarrow$ shrinks to 0 $\Rightarrow$ integral **converges**
- Positive exponent e^{at} , $a > 0 \Rightarrow$ grows without bound $\Rightarrow$ integral **diverges**

Example 3: Evaluate $\int_0^\infty e^{-st} dt$ for constant $s > 0$.

$$\int_0^T e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^T = -\frac{e^{-sT}}{s} + \frac{1}{s} \xrightarrow{T \rightarrow \infty} \frac{1}{s}$$

$$\boxed{\int_0^\infty e^{-st} dt = \frac{1}{s}, \quad s > 0}$$

We use this result repeatedly throughout the topic.

1.2 Integration by Parts — A Quick Refresher

$$\int u dv = uv - \int v du$$

Use the **LIATE** rule to choose u :

Priority	Type	Examples
1st	L ogarithms	$\ln t$
2nd	I nverse trig	$\arctan t$
3rd	A lgebraic	t, t^2
4th	T rigonometric	$\sin t, \cos t$
5th	E xponentials	e^{-2t}

Example: Evaluate $\int t e^{-st} dt$.

Choose $u = t$, $dv = e^{-st} dt$, so $du = dt$, $v = \frac{e^{-st}}{-s}$:

$$\int t e^{-st} dt = \frac{-t e^{-st}}{s} - \int \frac{e^{-st}}{-s} dt = \frac{-t e^{-st}}{s} - \frac{e^{-st}}{s^2} + C$$

We use this result when computing $\mathcal{L}\{t\}$.

2 Introduction — Why Do We Need Laplace Transforms?

2.1 The Problem with Our Current Methods

In Topics 2 and 3 we learned integrating factors, undetermined coefficients, and variation of parameters. These methods solve the equation first and apply initial conditions at the end, which can be long and error-prone for complicated forcing functions.

The Laplace Transform Strategy

ODE (in t) $\xrightarrow{\text{Step 1: Transform}}$ Algebraic equation (in s)

$\xrightarrow{\text{Step 2: Solve for } Y(s)}$ Answer in s -world

Step 3: Invert
—————→ Solution $y(t)$

Why is this better?

- Derivatives become simple algebra: y' becomes $sY - y(0)$
- Initial conditions are included automatically — not added at the end
- Complicated forcing functions are handled naturally

Analogy: To multiply $1,234 \times 5,678$ you could take logarithms, add, then convert back. Same idea — transform to a simpler world, work there, convert back.

3 Definition of the Laplace Transform

3.1 Building the Definition Step by Step

We multiply $f(t)$ by e^{-st} and integrate over $[0, \infty)$. The factor e^{-st} is a **damping weight**: for $s > 0$ it forces $e^{-st}f(t) \rightarrow 0$ as $t \rightarrow \infty$, ensuring convergence for a wide class of functions.

Definition 4.1 — The Laplace Transform

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$$

The integral must converge for the transform to exist.

Key Remarks

1. t disappears after integration — the result is a function of s only.
2. s is a new variable (the “frequency domain”). Think of it as a parameter.
3. **Capital letters for transforms:** $\mathcal{L}\{y(t)\} = Y(s)$, $\mathcal{L}\{x(t)\} = X(s)$, etc.
4. s must be large enough for the integral to converge.

3.2 Computing the Transform from the Definition — Four Core Examples

Example 1: Find $\mathcal{L}\{1\}$.

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st} dt = \frac{1}{s} \quad (\text{from Example 3 above})$$

$$\boxed{\mathcal{L}\{1\} = \frac{1}{s}, \quad s > 0}$$

Example 2: Find $\mathcal{L}\{e^{at}\}$.

$$\int_0^{\infty} e^{-st} \cdot e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt = \frac{1}{s-a} \quad (s-a > 0)$$

$$\boxed{\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s > a}$$

$f(t)$	a	$F(s)$	Valid when
e^{3t}	3	$\frac{1}{s-3}$	$s > 3$
e^{-5t}	-5	$\frac{1}{s+5}$	$s > -5$
e^{-t}	-1	$\frac{1}{s+1}$	$s > -1$

Example 3: Find $\mathcal{L}\{t\}$.

Using the integration by parts result from Section 1.2:

$$\mathcal{L}\{t\} = \lim_{T \rightarrow \infty} \left[\frac{-t e^{-st}}{s} - \frac{e^{-st}}{s^2} \right]_0^T$$

At the upper limit both terms $\rightarrow 0$ (exponential decay beats linear growth). At $t = 0$ both terms give $0 - (-1/s^2) = 1/s^2$:

$$\mathcal{L}\{t\} = \frac{1}{s^2}, \quad s > 0$$

General pattern:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad s > 0$$

Examples: $\mathcal{L}\{t^2\} = \frac{2}{s^3}, \quad \mathcal{L}\{t^3\} = \frac{6}{s^4}, \quad \mathcal{L}\{t^5\} = \frac{120}{s^6}.$

Example 4: Prove that evaluating the improper integral limits of the antiderivative yields the standard Laplace transform formula:

$$\mathcal{L}\{\sin bt\} = \frac{b}{s^2 + b^2}, \quad s > 0$$

Step-by-Step Derivation

1. *First Integration by Parts* Following the LIATE rule, Trigonometric (**T**) functions take priority over Exponential (**E**) functions when choosing u .

$$\begin{aligned} u &= \sin(bt) & dv &= e^{-st} dt \\ du &= b \cos(bt) dt & v &= -\frac{1}{s} e^{-st} \end{aligned}$$

Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$\begin{aligned} I &= -\frac{1}{s} e^{-st} \sin bt - \int \left(-\frac{1}{s} e^{-st} \right) (b \cos bt) dt \\ I &= -\frac{1}{s} e^{-st} \sin bt + \frac{b}{s} \int e^{-st} \cos bt dt \end{aligned} \tag{1}$$

Second Integration by Parts We apply integration by parts a second time to the remaining integral $\int e^{-st} \cos bt dt$, keeping the same assignment priorities.

$$\begin{aligned}
u &= \cos(bt) & dv &= e^{-st} dt \\
du &= -b \sin(bt) dt & v &= -\frac{1}{s} e^{-st}
\end{aligned}$$

Substitute this back into the new integral component:

$$\begin{aligned}
\int e^{-st} \cos bt \, dt &= -\frac{1}{s} e^{-st} \cos bt - \int \left(-\frac{1}{s} e^{-st} \right) (-b \sin bt) \, dt \\
\int e^{-st} \cos bt \, dt &= -\frac{1}{s} e^{-st} \cos bt - \frac{b}{s} \int e^{-st} \sin bt \, dt
\end{aligned}$$

Notice that our original integral I in (Equation 1) has reappeared inside this expression:

$$\int e^{-st} \cos bt \, dt = -\frac{1}{s} e^{-st} \cos bt - \frac{b}{s} I \tag{2}$$

Algebraic Isolation Substitute the result from Equation 2 into back into the Equation 1:

$$\begin{aligned}
I &= -\frac{1}{s} e^{-st} \sin bt + \frac{b}{s} \left(-\frac{1}{s} e^{-st} \cos bt - \frac{b}{s} I \right) \\
I &= -\frac{1}{s} e^{-st} \sin bt - \frac{b}{s^2} e^{-st} \cos bt - \frac{b^2}{s^2} I
\end{aligned}$$

Group all terms containing I on the left side of the equation:

$$I + \frac{b^2}{s^2} I = e^{-st} \left(-\frac{1}{s} \sin bt - \frac{b}{s^2} \cos bt \right)$$

Find a common denominator of s^2 for both sides to easily combine fractions:

$$\left(\frac{s^2 + b^2}{s^2} \right) I = e^{-st} \left(\frac{-s \sin bt - b \cos bt}{s^2} \right)$$

Multiply both sides by s^2 to eliminate the denominators:

$$(s^2 + b^2)I = e^{-st}(-s \sin bt - b \cos bt)$$

Finally, divide through by $(s^2 + b^2)$ to isolate I :

$$I = \frac{e^{-st}(-s \sin bt - b \cos bt)}{s^2 + b^2}$$

i Note

Adding an integration constant $+C$ completes the indefinite integral form.

$$\int e^{-st} \sin bt \, dt = \frac{e^{-st}(-s \sin bt - b \cos bt)}{s^2 + b^2} + C$$

$$\mathcal{L}\{\sin bt\} = \frac{b}{s^2 + b^2}, \quad s > 0$$

Given the antiderivative:

$$\int e^{-st} \sin bt \, dt = \frac{e^{-st}(-s \sin bt - b \cos bt)}{s^2 + b^2}$$

Set Up the Improper Integral By definition, the Laplace transform is an improper integral running from 0 to ∞ . We rewrite this using a formal limit with an upper bound $T \rightarrow \infty$:

$$\mathcal{L}\{\sin bt\} = \int_0^\infty e^{-st} \sin bt \, dt = \lim_{T \rightarrow \infty} \left[\frac{e^{-st}(-s \sin bt - b \cos bt)}{s^2 + b^2} \right]_0^T$$

Evaluate the Upper Limit ($t \rightarrow \infty$) We substitute T into the antiderivative and take the limit as $T \rightarrow \infty$:

$$\lim_{T \rightarrow \infty} \frac{e^{-sT}(-s \sin bT - b \cos bT)}{s^2 + b^2}$$

- **Exponential Behavior:** Under the constraint $s > 0$, the exponent $-sT$ remains strictly negative. As $T \rightarrow \infty$, the exponential term decays to zero: $\lim_{T \rightarrow \infty} e^{-sT} = 0$.
- **Trigonometric Behavior:** The terms $\sin bT$ and $\cos bT$ oscillate endlessly but are strictly bounded between -1 and 1 .
- **Product Rule for Limits:** A decaying exponential term multiplied by a bounded oscillating expression vanishes completely at infinity.

Upper Limit Value = 0

Evaluate the Lower Limit ($t = 0$) Next, we substitute $t = 0$ into the expression:

$$\text{Lower Limit Value} = \frac{e^{-s(0)}(-s \sin(b \cdot 0) - b \cos(b \cdot 0))}{s^2 + b^2}$$

Evaluating the elemental function values at 0: $e^0 = 1$ * $\sin(0) = 0$ * $\cos(0) = 1$

Substitute these values back into the fraction to simplify:

$$\text{Lower Limit Value} = \frac{1 \cdot (-s(0) - b(1))}{s^2 + b^2} = \frac{-b}{s^2 + b^2}$$

Apply the Fundamental Theorem of Calculus Subtract the lower limit value from the upper limit value (Upper – Lower):

$$\mathcal{L}\{\sin bt\} = 0 - \left(\frac{-b}{s^2 + b^2} \right)$$

Simplifying the double negative yields the final proof:

$$\boxed{\mathcal{L}\{\sin bt\} = \frac{b}{s^2 + b^2}, \quad s > 0}$$

Your Turn: Prove

$$\mathcal{L}\{\cos bt\} = \frac{s}{s^2 + b^2}, \quad s > 0$$

Note: $\sin bt$ has numerator b ; $\cos bt$ has numerator s .

$f(t)$	b	$F(s)$
$\sin 3t$	3	$\frac{3}{s^2 + 9}$
$\cos 3t$	3	$\frac{s}{s^2 + 9}$
$\sin 7t$	7	$\frac{7}{s^2 + 49}$
$\cos t$	1	$\frac{s}{s^2 + 1}$

4 Standard Transforms Table

#	$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$	Condition
1	1	$\frac{1}{s}$	$s > 0$
2	t	$\frac{1}{s^2}$	$s > 0$
3	t^n	$\frac{n!}{s^{n+1}}$	$s > 0$
4	e^{at}	$\frac{1}{s - a}$	$s > a$
5	$\sin bt$	$\frac{b}{s^2 + b^2}$	$s > 0$
6	$\cos bt$	$\frac{s}{s^2 + b^2}$	$s > 0$
7	$\sinh bt$	$\frac{b}{s^2 - b^2}$	$s > b $
8	$\cosh bt$	$\frac{s}{s^2 - b^2}$	$s > b $
9	$t^n e^{at}$	$\frac{n!}{(s - a)^{n+1}}$	$s > a$
10	$e^{at} \sin bt$	$\frac{b}{(s - a)^2 + b^2}$	$s > a$
11	$e^{at} \cos bt$	$\frac{s - a}{(s - a)^2 + b^2}$	$s > a$



Tip

Using the table:

- **Forward transform:** find $f(t)$ in the left column, read $F(s)$ in the middle.
- **Inverse transform:** find $F(s)$ in the middle column, read $f(t)$ on the left.